

Bixuan LIU

☎ +33 06 85 04 56 67 ✉ bixuan.liu@sorbonne-universite.fr 📍 Paris, France
🌐 www.linkedin.com/in/bixuan-liu-008b732b9 💻 <https://bi-xuan.github.io/>

PROFILE

PhD student at the Laboratoire de Probabilités, Statistique et Modélisation (LPSM), Sorbonne University, Paris, France, specializing in **statistical and machine learning methods** to reconstruct networks in ecological and biological systems. My research combines mathematical modelling, statistical inference, and complex data analysis. I'm currently seeking opportunities in **data science** across the financial services, pharmaceutical, healthcare, and technology sectors.

EDUCATION

- Sorbonne Université** Top100 Sep 2024 - Present
PhD in Statistical Modeling Paris
Laboratoire de Probabilités, Statistique et Modélisation (LPSM)
• Supervisors: Stéphane Robin (LPSM, Paris), Viet Chi Tran (Inria, Lille), and Elisa Thébault (iEES, Paris)
- University of Luxembourg** Sep 2022 - Aug 2024
Master in Data Science Luxembourg
• GPA: 17.0 / 20.0
• Relevant Coursework: Probabilistic Models in Finance (20/20), Mathematical Statistics (19/20), Big Data Analytics (18/20), Reproducible Pipelines and Time Series Forecasting (18/20), Prototyping with Deep-Learning (18/20)
- Southeast University** 985 211 Sep 2019 - Jun 2023
Bachelor in Mathematics and Applied Mathematics Nanjing
• GPA: 3.68 / 4.00
• Relevant Coursework: Mathematical Statistics (98/100), Optimization Method (92/100), Stochastic Processes (98/100), Mathematical Analysis II (90/100), Programming and Algorithmic Language I (90/100)

RESEARCH EXPERIENCE (SELECTED)

- NeurIPS 2026 Virtual Embryo Challenge (ongoing)** Sep 2026 - Present
Objective: Reconstruct and predict embryonic development
• Training a diffusion model that jointly captures spatial context, temporal dynamics, biological hierarchies, and responses to perturbations
- Dynamical Interaction Network Reconstruction** Jun 2025 - Present
Sorbonne Université Paris
Github repository: <https://github.com/Bi-xuan/MS-S.git>
• Developed an estimator based on model selection technique with theoretical guarantees on optimal risk
• Designed a scalable, heuristic-based algorithm to reconstruct interaction network through simulated data, addressing a setting with no established methods and implementing on a high-performance computing (HPC) cluster
• Adapted the method to a real-world ecological dataset of rodent-flea networks comprising more than 100 species, identifying ecologically plausible interactions.
• Extended the framework to large-scale network reconstruction using reinforcement learning
- System Identification for Time Series Models** Jan 2024 - Jun 2025
Sorbonne Université Paris
• Established identifiability conditions for first-order vector autoregressive (VAR(1)) systems using only stationary statistics through algebraic-statistical methods.
- International Fintech Innovation Competition** Aug 2021 - Sep 2021
Team Leader
• Led a team and was responsible for data analysis, data processing, and solution design; the team ranked in the top 36 nationwide

PUBLICATION

- Identifiability of VAR(1) in a stationary setting** Apr 2025
Bixuan Liu
Submitted to Electronic Journal of Statistics (EJS)
• Preprint: <https://arxiv.org/abs/2504.03466>
- Support recovery in stationary VAR(1) models via model selection** Present
Bixuan Liu, Stéphane Robin, Viet Chi Tran

PROJECTS

Reproducible Data Analysis Pipeline for Luxembourg Labor Market

Jan 2024 - Jan 2024

University of Luxembourg

Luxembourg

GitHub repository: https://github.com/Bi-xuan/Project_RAP.git

- Developed a reproducible analytical pipeline in R using Luxembourg labor market data, including data visualization, unemployment rate calculation, and regression modeling

Music Generation with Deep Learning

Sep 2023 - Dec 2023

University of Luxembourg

Luxembourg

- Developed transformer-based models for music generation, producing coherent sequences
- Implemented training pipelines in Python with HPC cluster for parallel training

MRI Tumor Detection with CNN

Jan 2022 - Feb 2022

2022 Data Science Winter School at Imperial College London

Remote

- Developed CNN models for glioma detection from medical images, achieving 85% accuracy on the validation set
- Received an A grade in the overall project assessment

ACADEMIC ACTIVITIES

"Identifiability of VAR(1) in a Stationary Setting"

Presented at:

- Journées de Statistique, Graphs and Networks session, *Clermont-Ferrand* — Jun. 1, 2026
- Optimization Seminar, *Bordeaux* — Apr. 16, 2026
- Doctoral Seminar of LPSM, *Paris* — Apr 7, 2026
- Algebraic Statistics Workshop, *Lille* — Mar 13, 2026
- Seminar on Random Modeling of Living Systems, *Paris* — Oct 1, 2025
- Summer School of the Chaire MMB, *Aussois* — Jun. 19, 2025

TEACHING MISSIONS

Teaching Assistant

Sep 2025 - Present

Sorbonne Université

Paris

- Led tutorials in: Analysis of Multivariate Data, Mathematics for Science II: Algebra and Probability, Probability
- Conducted tutorial sessions, designed exercises, and graded coursework

EXTRACURRICULAR ACTIVITIES

2022 COMAP's Mathematical Contest in Modeling

Feb 2022 - Feb 2022

- Responsible for modeling and thesis writing; the team received an Honorable Mention

Orientation Volunteer Team

Sep 2020 - Sep 2020

- Provided support to incoming students, including distributing food and drinks and answering questions

Student Union in School of Mathematics

Sep 2019 - Aug 2021

Minister Literature and Art Department

- Organized the Orientation Party and Annual Commendation Conference for the School of Mathematics, securing sponsorship of RMB 20,000 and attracting thousands of participants
- Hosted a cultural event, "Night Talks Around the Furnace," recognized as an excellent activity by the School of Mathematics

HONORS & AWARDS

Student Scholarship of University of Luxembourg

2022-2023 Academic Year

Certificate of Excellence

2022-2023 Academic Year

Individual Award (Excellent Social Work Award)

2020-2021 Academic Year

Course Award

2020-2021 Academic Year

MISCELLANEOUS

- Digital Tools:** Office (Powerpoint, Excel), Adobe Photoshop
- Programming & Data Analysis:** Python, R, Matlab, Java, SQL, NoSQL
- Languages:** English(TOEFL iBT:104), French(B2), Chinese(native)
- Interests:** Badminton, Reading, Traveling, Swimming